Winter Storm Fern Situation Report - January 2026

Formatted based on LA Fire Situation Report

Summary

A major American winter storm, often referred to as Winter Storm Fern, started on Friday, January 23 and continued through January 26th. The storm brought heavy snow and freezing rain to several U.S. states, ranging from the southern plains to the Eastern United States. Several states experienced network outages and extremely cold temperatures.

The following visualizations are developed based on Meta AI for Good's data through Facebook. The data covers network coverage and population density from January 27 to February 10, with a focus on Tennessee and Kentucky and daily updates during the collection period.

Movement

% Population Density Change

Note the difference in Figure 1, which is based on Bing tiles, and Figure 2, which is based on administrative regions.

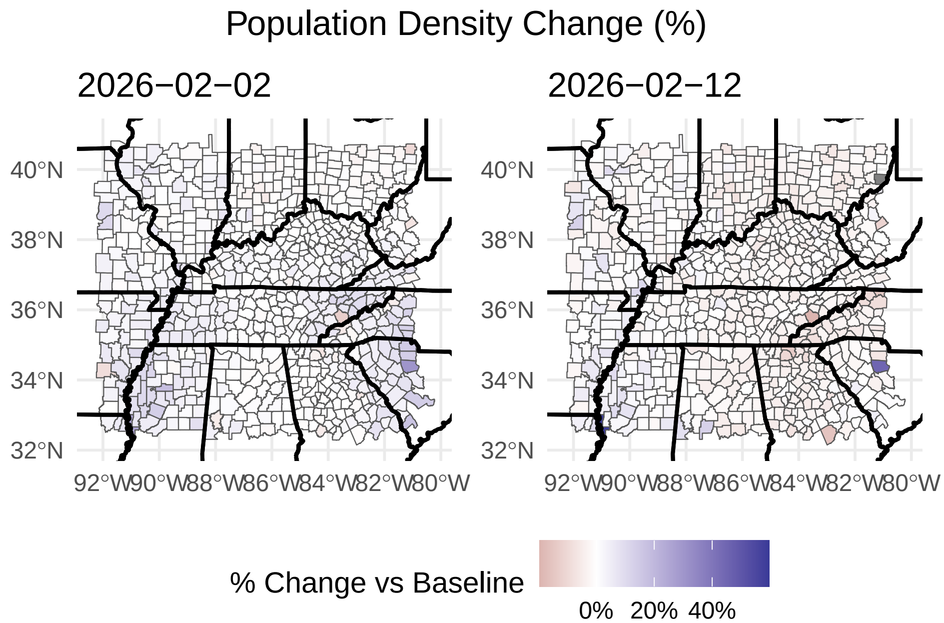

Figure 1: Based on the Bing tiles, there are only a few counties with strong population density changes.

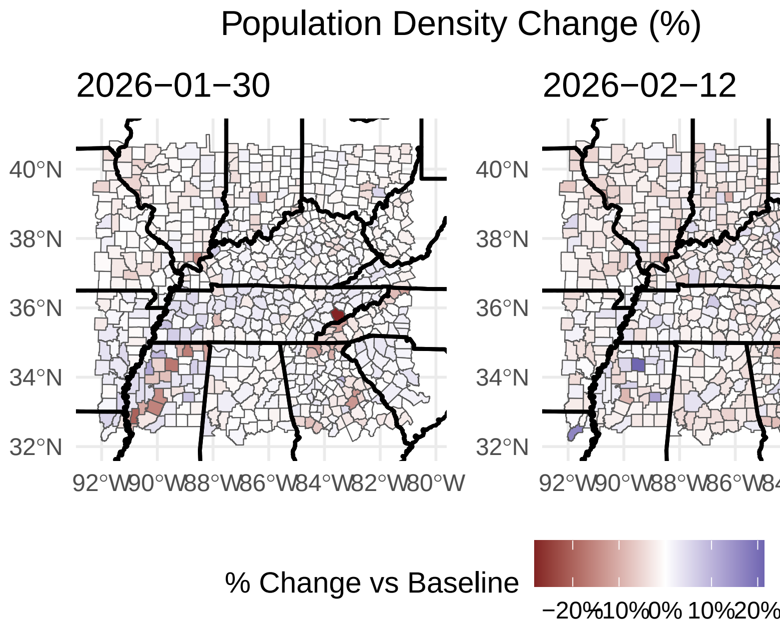

Figure 2: Based on the administrative regions, there are two counties with strong population density changes, one positively and one negatively.

Time Series of Movement

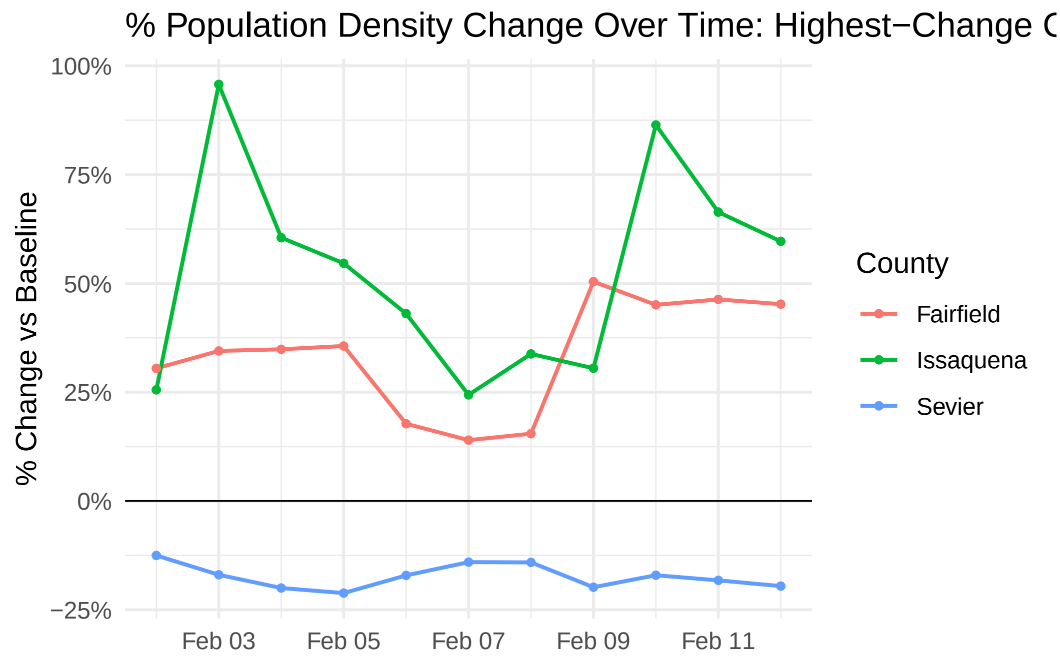

Figure 3: There are strong difference in population changes across different counties

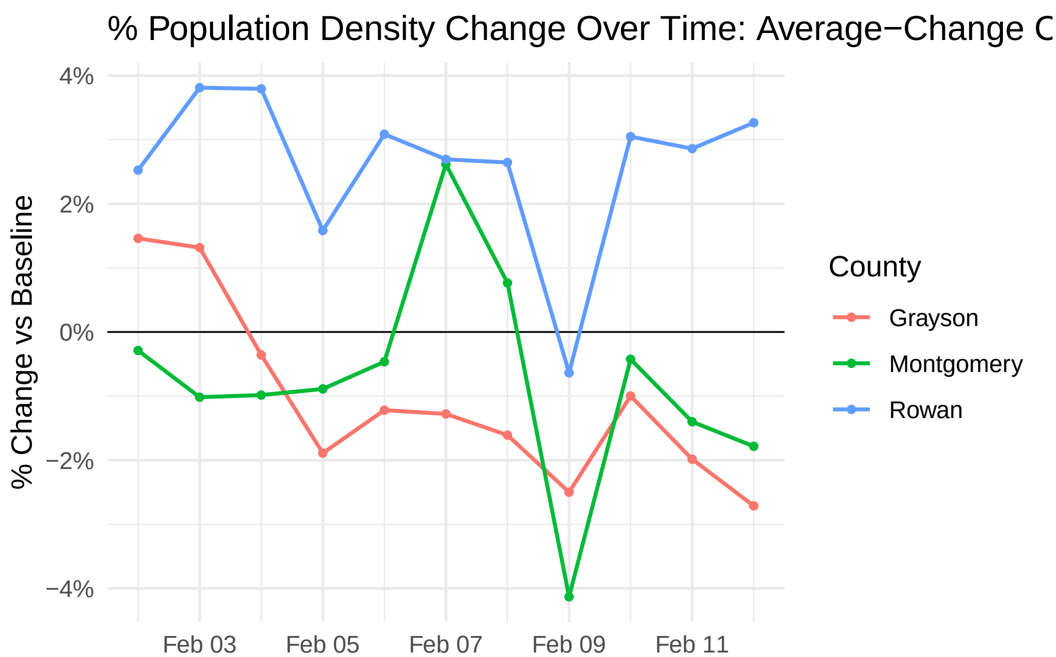

Figure 4: There are strong difference in population changes across different counties

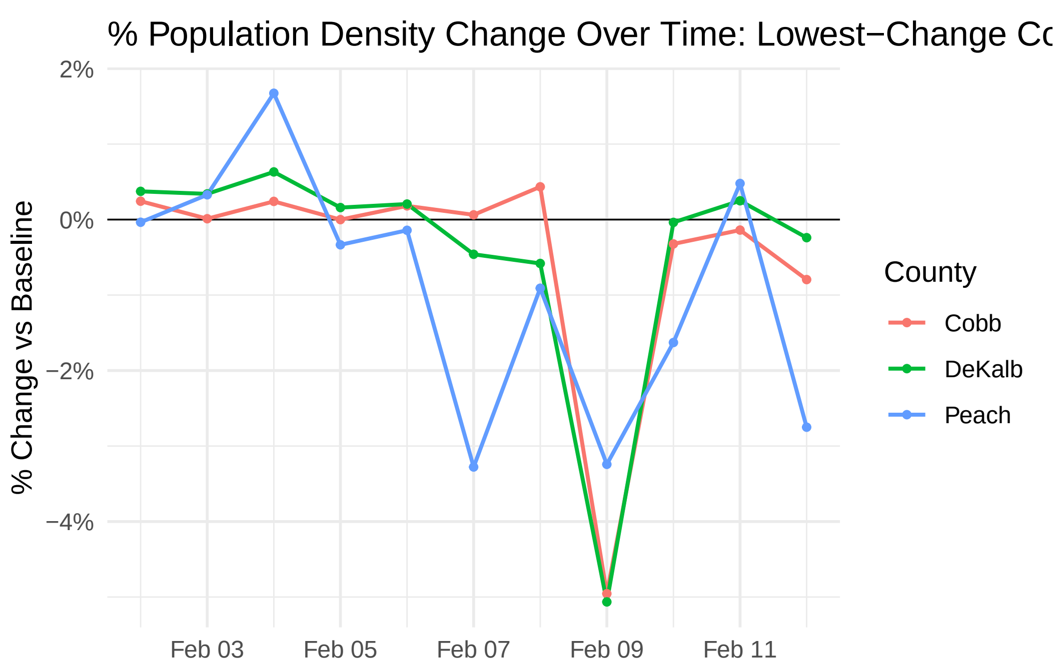

Figure 5: There are strong difference in population changes across different counties

Check why lowest tracks so closely on Feb 9th and investigate whether outliers are skewing mean

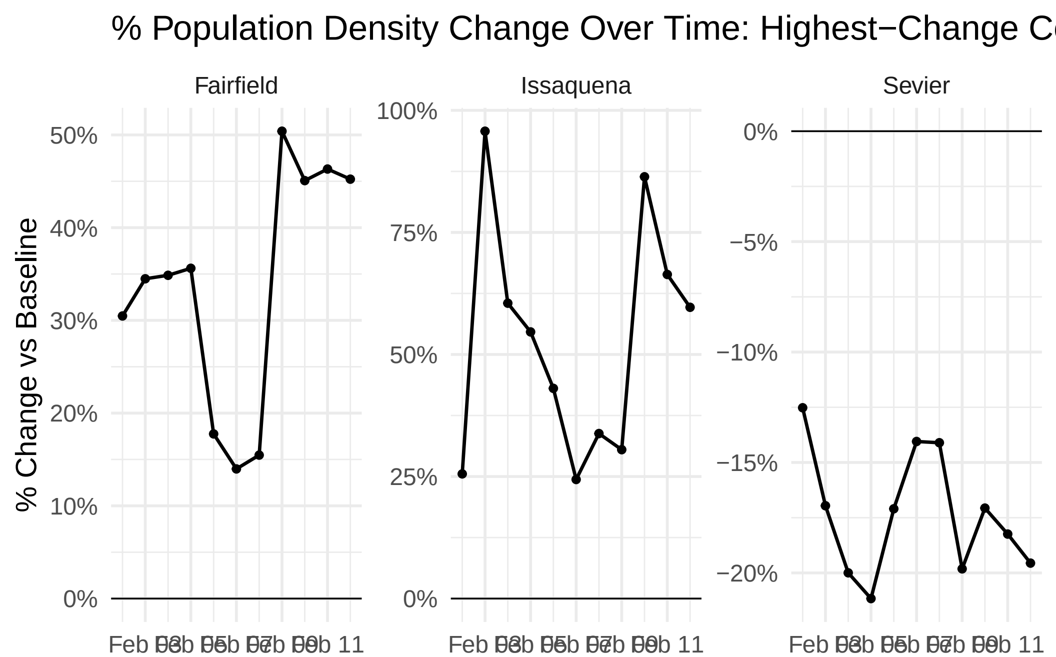

Figure 6: There are strong difference in population changes across different counties

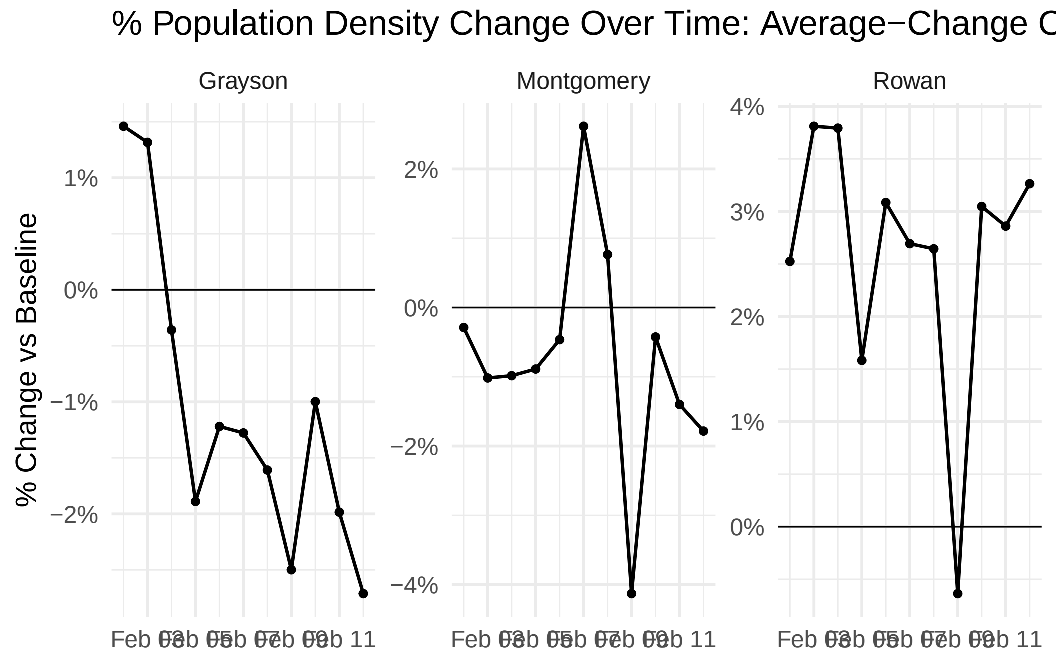

Figure 7: There are strong difference in population changes across different counties

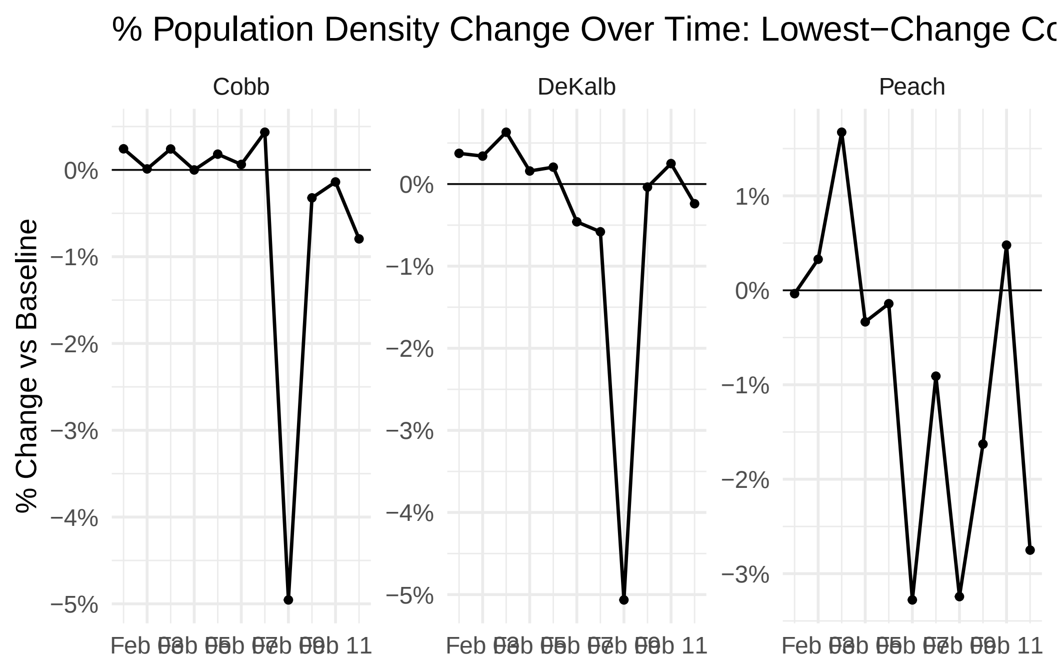

Figure 8: There are strong difference in population changes across different counties

Table 1

start_county_name	end_county_name	total_crisis_movement	avg_z_score
Sevier	Russell	0	NaN
Shelby	Woodruff	0	NaN
Smith	Sevier	0	NaN
Spartanburg	Cocke	0	NaN
Sumner	Marion	0	NaN
Swain	Greenville	0	NaN
Tippecanoe	Davidson	0	NaN
Tuscaloosa	Cherokee	0	NaN
Vanderburgh	McCracken	0	NaN
Walker	DeSoto	0	NaN
Warren	Fulton	0	NaN
Watauga	Knox	0	NaN
Weakley	Lake	0	NaN
White	Sevier	0	NaN
Williamson	Gibson	0	NaN
Williamson	Gwinnett	0	NaN
Wood	Fairfield	0	NaN
Wood	Hocking	0	NaN
Wood	Tuscarawas	0	NaN
Wythe	Raleigh	0	NaN

Places

As shown by Table 1, …

Demographics

Used sample measures that I thought said more than the normal ones. Easily changed in the function so if you see one that's cool lmk

Not finished but you get the gist. I also made it so i can pick which day the population density change is from (in progress)

County	State	Population Density Change	Total Population	Median Household Income	Poverty Rate	% Age 65+	% No Vehicle Access
Issaquena	Mississippi	59.7	928	$31,429	16.2%	2.7%	14.0%
Fairfield	South Carolina	45.2	20,550	$47,885	15.6%	3.2%	5.9%
Sevier	Tennessee	-19.6	99,652	$62,581	13.5%	2.7%	4.1%
Laurens	Georgia	-15.6	49,798	$55,010	22.6%	2.3%	6.3%
Fannin	Georgia	-13.8	25,742	$57,073	13.7%	2.9%	3.0%
Alleghany	North Carolina	-13.8	11,174	$47,172	16.0%	2.6%	6.0%
Osage	Missouri	12.6	13,402	$76,681	8.6%	2.6%	5.6%
Perry	Alabama	12.6	8,031	$37,654	31.4%	3.3%	8.0%
Clay	West Virginia	-11.3	7,835	$42,318	26.4%	3.5%	10.3%
New Madrid	Missouri	10.6	15,731	$51,881	17.7%	3.3%	7.2%